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Policy fundamentals, interest rates differential, and expected devaluation in the presence of an active crawling peg system

GRACIANA DEL CASTILLO*

Office of the Director General for International Economic Co-operation and Development, United Nations and Columbia University, New York, USA

One of the most interesting theoretical issues that comes up from the analysis of the *tablita* period in the Southern Cone economies—Argentina, Chile and Uruguay—is the interpretation of why interest rates on domestic currency loans were much higher than the rate on foreign currency loans adjusted for devaluation of the domestic currency, even though the rate of devaluation was set by the monetary authorities six months ahead and published in a little table (*tablita*). This was particularly puzzling in the case of Uruguay which had practically perfect capital mobility. We show that the reason rational economic agents were willing to pay high domestic interest rates rather than borrowing in foreign currency was that they realized that policy fundamentals were inconsistent with the sustainability of the exchange rate policy. Using data for Uruguay we show that a second systematic component of expectations of devaluation is what we call 'the policy risk factor' and modeled as a function of the fiscal and monetary variables. (JEL 41, 31).

In spite of the many research efforts to model interest rate determination, no model has yet been able to explain the extraordinarily high interest rates following financial liberalization programs in the Southern Cone economies of Latin America—Argentina, Chile, and Uruguay—in the 1970s. In the late 1970s these countries had adopted what became known as the *tablita*—a managed exchange rate system in which the future daily path of the exchange rate was preannounced for several months ahead.[1]

One of the most interesting issues that comes up from the analysis of the *tablita* period is the interpretation of why nominal domestic interest rates were much higher than foreign rates adjusted for currency devaluation as scheduled in the *tablita*. This was particularly puzzling in the case of Uruguay which, following financial liberalization in the mid-1970s became an economy fully integrated through the capita account (with no restrictions whatsoever). In fact, as a result

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of financial reforms Uruguay had become, by the second half of the 1970s, one of the least restricted financial markets in the world.[2]

The purpose of this paper is to model the interest rate differential between interest rates in domestic currency and interest rates in foreign currency in the domestic economy after adjusting for devaluation of the exchange rate (currency adjusted spread). The model differs from other models of this kind in the way we shall model the expected rate of devaluation of the domestic currency. The paper is divided as follows: in Section I we discuss what we believe are the drawbacks of alternative models to explain interest rates in small open economies with a *tablita* system, but in which there is uncertainty about the continuation of the active crawling peg system. In Section II we present our model and test it using data for Uruguay. The selection of Uruguay for purpose of analysis was made for two main reasons. First, Uruguay provides a classic example of the well-known 'small-country' assumption. Second, as we pointed out earlier, throughout the period of analysis there were no restrictions whatsoever on capital flows.[3] In Section III we summarize the main results and draw some conclusions.

I. Interest rate determination: alternative models

The basic problem with interest rate determination models in the presence of a *tablita* exchange rate system which is not fully credible is how to model expectations of devaluation, particularly during periods in which there is high uncertainty about the sustainability of the foreign exchange policy.

Edwards and Khan (1985) contend that most developing countries are neither completely 'open' nor completely 'closed' and therefore they develop a framework for interest rate determination in which interest rates depend on world interest rates and on the expected rate of devaluation as well as on domestic market solutions (unstarred variables relate to the home country while starred variables relate to the foreign country, standing proxy for the 'rest of the world'):

$$\langle 1 \rangle \qquad i_t = \psi(i_t^* + \delta_t^e) + (1 - \psi)(r_t + \pi_t^e),$$

where i_t is the nominal domestic rate of interest, i_t^* is the world rate of interest, δ_t^e is the expected rate of devaluation, r_t is the real rate of interest, and π_t^e is the expected rate of inflation. The parameter ψ measures the degree of financial openness of the country: the closer ψ is to one, the more open the economy is and the larger the weight of foreign factors in interest rate determination.[4] They specify the real rate as

$$\langle 2 \rangle \qquad r_t = \kappa - \lambda EMS_t + \omega_t,$$

where κ is a constant (representing long-run equilibrium real interest rate), EMS represents the excess supply of money,[5] λ is a parameter ($\lambda > 0$), and ω_t a random error term.

The solution for *ex ante* domestic nominal interest rates in a semi-open economy is thus[6]

$$\langle 3 \rangle \qquad i_t = \psi(i_t^* + \delta_t^e) + (1 - \psi)(\kappa - \lambda EMS_t + \pi_t^e + \omega_t).$$

To estimate equation $\langle 3 \rangle$, however, assumptions have to be made regarding unobservable variables such as EMS_t,[7] π_t^e,[8] and δ_t^e. In the case of the Southern

Cone economies during the *tablita* period, it is very hard to determine δ_t^e using conventional models.[9]

We could postulate that the expected rate of change of the exchange rate follows an adaptive expectations model where expectations of a devaluation are assumed to follow a geometrically distributed function of past rates of devaluation of the exchange rate. During the experiment with the *tablita* exchange rate system in the Southern Cone economies, this formulation would not pick up the effect of the changes in expectations as the probability of a change in the exchange rate regime, and therefore the spread between the peso and dollar rates of interest, increased in these economies over time.

If we assume rational expectations in which economic agents have full information and know the model, and, in particular, if we assume perfect foresight where expectations of exchange rate depreciation are given by the actual rate as specified in the *tablita* schedule ($\delta_t^e = \delta_t^t$), we would grossly underestimate expectations of exchange rate changes since the actual rate of devaluation (as scheduled in the *tablita*) was overall very much lower than what economic agents expected the rate of change in the exchange rate to be.

In equations estimated for Colombia by Edwards and Khan (1985), the expected rate of devaluation between periods t and $t + 1$ was replaced by the actual rate of devaluation in period t (on the assumption that the rate of devaluation could be represented approximately as a random walk process with zero drift). This would not be a reasonable assumption in the presence of a *tablita* where the rate of devaluation was preannounced.

If economic agents are assumed to be risk neutral, the expected rate of change of the exchange rate would be equal to the forward exchange premium. Because forward rates are available for Singapore, Edwards and Khan used them to proxy the expected change in the exchange rate in that country. Although there was no forward market as such in the economies of the Southern Cone, during the *tablita* period the forward premium was *de facto* given by the *tablita* schedule. In this case (just as when we assumed perfect foresight), expectations of devaluation were much higher than the scheduled rate.

Edwards and Khan (1985) point out that an alternative way to deal with the risk factor is to explicitly introduce a risk premium into the analysis. They do so by assuming that the risk premium is equal to a constant plus a random term. The constant part of the premium is added to the constant in the equation they estimate, while the random component becomes part of the error term. We will posit that this modeling of the expected rate of devaluation is unsatisfactory in the case of interest rate determination in the Southern Cone economies during the *tablita* period. With no futures markets as such, expectations about the sustainability of the exchange rate regime were of crucial importance in the formation of expectations about the rate of devaluation. Such expectations determined the premium that creditors charged on their loans in domestic currency and the premium borrowers were willing to pay on their loans, over and above the dollar rate of interest adjusted for the rate of devaluation as scheduled in the *tablita*.

Hanson and de Melo (1985) developed a model of uncovered interest parity to explain the interest rate differential in Uruguay between identical financial instruments with different currency denominations. Letting i_t^p and i_t^d be the annualized interest rates on peso and dollar one–six months deposits, respectively, we can express their model as

$$\langle 4 \rangle \qquad i_t^p = i_t^d + \delta_t^e + \mu_t,$$

where δ_t^e is the expected rate of devaluation proxied by the annualized *ex post* rate of devaluation for six months.

Hanson and de Melo tested their model using monthly data for Uruguay for the period November 1978 to December 1981. Thus, they excluded the period of greatest foreign exchange uncertainty prior to the November 1982 devaluation.[10] Nevertheless, although they found that nearly 90 per cent of the variance around a constant spread could be explained by the included variables, the *ex post* rate of devaluation was found to be insignificant.[11] As they point out, this result indicates that changes in the *ex post* rate of devaluation, if preannounced, had no effect on the *ex ante* spread. They find this result 'surprising'. As we will discuss below, while the *tablita* was in operation in Uruguay, the actual rate of devaluation as scheduled in the *tablita* was an inadequate descriptor of expectations of devaluation.

Their model is misspecified since it does not explicitly take into account uncertainty about the probability that the foreign exchange regime would be discontinued. As we said earlier, uncertainty about government policy was a crucial factor in the formation of expectations of devaluation (particularly in the latter period of the *tablita*). Hence, the way they model expectations is even less adequate for 1982 when expectations about the unsustainability of the exchange rate parity had increased.[12]

More recently, Connolly and Taylor (1990) also question the interest arbitrage equation under a *tablita* exchange rate system such as Argentina had in 1979–81 and later on under the Austral Plan (June 1985 to April 1986) and the Plan Primavera (August 1988 to January 1989). They argue that the Austral is subject not only to the preannounced rate of devaluation but also to the additional risk of a deviation from the planned rate by a surprise devaluation. They posit that the unplanned devaluation component is in fact known since individuals know by how much the currency has appreciated beyond the equilibrium level. They assume that the extent of real appreciation is easily calculated and widely available so that investors reason that the authorities will ultimately restore the equilibrium level, but it is not known when they will do so. In their model there is certainty regarding the rate of unplanned devaluation, but uncertainty regarding its timing. A risk premium is built into the interest rate so as to compensate for the additional risk involved in holding Australes. The risk premium depends on the degree of the investor's risk aversion, as embodied in the shape of the investor's utility function.

II. Our model

The purpose of the model is to explain the interest rate differential between similar financial instruments with different currency denominations after adjusting for changes in the exchange rate (currency adjusted spread), in small, dual currency, inflationary economies that have adopted a *tablita* type of exchange rate system—but in which there is uncertainty about the government and its policies.

The advantage of modeling domestic interest rates in relation to foreign interest rates in the domestic economy rather than in relation to world interest rates (i_t^*)

is that it allows us to abstract from differences between assets across countries and allows us to assume that assets are perfect substitutes in the country (that is, that their prices move up or down more or less in line together). During the *tablita* period, economic agents had the clear choice of borrowing (or lending) in pesos or in dollars and whether they did one or the other depended exclusively on their expectations about the future path of the exchange rate. As we said earlier, with no futures markets as such, expectations about the sustainability of the exchange rate policy were of crucial importance in the way rational economic agents formed their expectations about the rate of devaluation. Such expectations determined the premium that rational borrowers were willing to pay on their loans, over and above the dollar rate of interest adjusted for the rate of devaluation as preannounced in the *tablita*.

Letting the peso be the domestic currency and the dollar the foreign currency, we formulate the spread between the nominal peso rate of interest (i_t^p) and the dollar rate of interest (i_t^d) as equal to the expected rate of devaluation of the domestic currency (δ_t^e)

$$\langle 5 \rangle \qquad i_t^p - i_t^d = \delta_t^e.$$

The model differs from other models of this kind in the way we shall model the expected rate of devaluation of the domestic currency. To explain the interest rate puzzle during the *tablita* period it is important to separate the rate of devaluation into three parts: one which we shall call the '*tablita* factor', that is, the actual devaluation of the peso as preannounced in the *tablita* schedule (δ_t^t);[13] a second component, which is also systematic and we shall call the 'policy risk factor' (δ_t^r); and an error term (ε_t) representing not only political risks that are not systematic but all external shocks which are stochastic. Thus, the devaluation of the domestic currency is expressed as

$$\langle 6 \rangle \qquad \delta_t = \delta_t^t + \delta_t^r + \varepsilon_t.$$

The expected value of the conditional expectation of ε_t given the policy variables (that we shall specify below) is zero

$$\langle 7 \rangle \qquad E[\varepsilon_t/\text{Policy var.}] = 0.$$

The expected rate of devaluation can be written as

$$\langle 8 \rangle \qquad \delta_t^e = \delta_t^t + E(\delta_t^r).$$

The experience of the Southern Cone economies indicates that, in spite of the *tablita*, expectations of devaluation during the period in which the rate of devaluation was preannounced were much higher than the rate determined by the schedule. Looming in the background was the fact that governments would change the existing exchange rate regime, either by changing the base of the *tablita*, by eliminating the *tablita* altogether, or by imposing exchange controls. Equally important was the risk that economic policies would change either as a result of a change in government or as a result of a change in the economic team. There were no explicit contracts throughout the *tablita* period to cover for these risks.[14]

We assume, *ex ante*, that the expected rate of devaluation over and above what is specified in the *tablita* schedule is equal to the conditional expectation of the policy risk factor given the policy variables

$$\langle 9\rangle \qquad \delta_t^e - \delta_t^t = E[\delta_t - \delta_t^t/\text{Policy var.}],$$

$$\langle 10\rangle \qquad = E[\delta_t^r/\text{Policy var.}].$$

We also assume that the conditional expectation is linear

$$\langle 11\rangle \qquad \approx \gamma Z_t,$$

where Z_t is a function of fiscal and monetary variables which economic agents take into account when they are forming their expectations about the sustainability of government policies.

In particular we postulate that Z_t is basically a function of what is happening to the government budget deficit in relation to GDP (G_t) and to the international reserves of the Uruguayan banking system (TRU_t). Given the fact that the Uruguayan economy is very much dependent on and integrated with Argentina,[15] we postulate that Z_t is also a function of international reserves in Argentina (TRA_t) and domestic interest rates in that country (IA_t),[16] since expectations in Uruguay are highly dependent on political and economic events in Argentina.

$$\langle 12\rangle \qquad Z_t = Z_t[G_t, TRU_t, TRA_t, IA_t].$$

Thus, while we stated that the formation of expectations during the *tablita* period was not consistent with assumptions of perfect foresight, our model is nevertheless consistent with rational expectations since we assume economic agents have full information and make use of the economic model in forming their expectations.

Since the policy risk factor is equal to

$$\langle 13\rangle \qquad \delta_t^r = \gamma Z_t + \eta_t, \qquad \text{where } E(\eta_t) = 0,$$

from equations ⟨6⟩ and ⟨13⟩, the difference between the actual rate of devaluation and the devaluation rate specified in the *tablita* is equal to

$$\langle 14\rangle \qquad \delta_t - \delta_t^t = \delta_t^r + \varepsilon_t = \gamma Z_t + \eta_t + \varepsilon_t.$$

From equations ⟨5⟩, ⟨6⟩, ⟨8⟩, and ⟨12⟩, we explain deviations from interest rate parity as follows

$$\langle 15\rangle \qquad i_t^p - i_t^d - \delta_t^t = D_t = E(\delta_t^r) = \gamma Z_t,$$

where D_t can be thought of as the 'currency adjusted spread.'[17]

After substituting for Z_t in equation ⟨15⟩ we estimated this equation for the *tablita* period (1978:10 to 1982:11)[18] with variables expressed as three-month moving averages and where G_t is the share of the budget deficit to GDP_t;[19] TRU_t and TRA_t are percentage changes in total international reserves in Uruguay and Argentina,[20] respectively; and IA_t is the interest rate in Argentina.[21]

We tested for unit roots (non-stationarity) in the dependent and independent variables by running an Augmented Dickey–Fuller (ADF) regression for each variable of the following type

$$\langle 16\rangle \qquad X_t - X_{t-1} = \beta_0 X_{t-1} + \beta_1 \Delta X_{t-1} + \beta_2 \Delta X_{t-2}.$$

The null hypothesis is that $\beta_0 = 0$, that is, the variable is a random walk (see Table 1).

TABLE 1. Unit root tests for the variables of interest.

Variable	Name	t-statistic
1	D	−1.98
2	G	−1.11
3	TRU	−3.68
4	TRA	−2.67
5	IA	−1.78
Critical values (50 observations)*		
1 per cent	5 per cent	10 per cent
−3.61	−3.16	−2.92

Source: Guilkey and Schmidt (1989).

Second, we took the residuals from the estimated regression and did an ADF test on them as follows

$$\langle 17 \rangle \qquad \hat{U}_t - \hat{U}_{t-1} = \beta_0 \hat{U}_{t-1} + \beta_1 \Delta \hat{U}_{t-1} + \beta_1 \Delta \hat{U}_{t-2}.$$

The null hypothesis is that $\beta_0 = 0$ or that the error term from the regression has a unit root and the variables are not co-integrated.

With 47 observations, the t-statistic on β_0 was -3.85. The critical values with 50 observations and five variables in the co-integrating equation are -4.8 (1 per cent), -4.15 (5 per cent), -3.85 (10 per cent). We cannot reject the hypothesis that we have a unit root in the residuals at the 95 per cent confidence level.

Because of the time series properties of the variables the appropriate specification is in first differences rather than in levels. Letting $\Delta D_t = D_t - D_{t-1}$, LS results were as follows[22]

$$\langle 18 \rangle \qquad \begin{array}{l} \Delta D_t = \underset{(0.21)}{0.81} + \underset{(3.22)}{0.48}\Delta G_t - \underset{(-2.93)}{0.24}\Delta TRU_t - \underset{(-2.40)}{0.17}\Delta TRA_t - \underset{(-2.32)}{0.11}\Delta IA_t, \\ R^2 = 0.77 \quad \bar{R}^2 = 0.74 \quad \mathrm{DW} = 1.84 \quad F = 24.07 \quad AR = 2^{23} \end{array}$$

where D_t is the *ex post* currency adjusted spread because the adjustment was made with δ_t^e given by the *ex post* actual depreciation of the peso.[24]

All coefficients are significant.[25] The first three coefficients have the predicted sign. An increase in the budget deficit in relation to GDP increases the *ex post* currency adjusted spread in the same way as a fall in international reserves, not only in Uruguay but in Argentina as well. As regression ⟨18⟩ indicates, changes in the currency adjusted spread and changes in the budget deficit move in the same direction while changes in the spread and changes in international reserves both in Uruguay and Argentina move in the opposite one.

We made no *a priori* assumption however on how interest rates in Argentina would affect the *ex post* currency adjusted spread in Uruguay. On the one hand, increasing interest rates in Argentina could lead Argentinians to borrow in Uruguay, raising interest rates there. We found, however, that changes in interest rates in Argentina had a negative impact on changes in the spread in Uruguay which indicates that a fall in interest rates in Argentina increases the spread in Uruguay. This could be a sign that during periods of declining interest rates in Argentina, there are capital outflows some of which are directed to Uruguay. Given the very small foreign exchange market, inflows of dollars from Argentina seem to put more downward pressure on dollar interest rates and on the value of the dollar than on domestic interest rates, thus also increasing the currency adjusted spread in Uruguay.

These results indicate that fiscal and monetary variables were successful in explaining the *ex post* currency adjusted spread during the *tablita* period. The growing fiscal deficit, the fall in international reserves both in Uruguay and Argentina, and interest rates in Argentina are able to explain the *ex post* devaluation of the peso in 1982. On the other hand, these variables could not explain the *ex ante* currency adjusted spread when the *tablita* factor (δ_t^t) was given by the devaluation of the peso as preannounced in the *tablita* schedule. The reason is that fiscal and monetary variables were obviously inconsistent with the maintenance of the *tablita* schedule during the latter part of the period, and were therefore the 'cause' of the 1982 devaluation of the peso.

III. Conclusions

While the *tablita* exchange rate system was in operation in Uruguay, the rate of devaluation as scheduled in the *tablita* was an inadequate descriptor of expectations of devaluation. A second systematic component of expectations of devaluation was what we called the 'policy-risk factor' and modeled as a function of the fiscal imbalance, international reserves both in Uruguay and Argentina, and interest rates in the latter country. These fiscal and monetary variables were found to be successful in explaining the *ex post* but not *ex ante* currency adjusted spread between peso and dollar rates of interest. In other words, the fiscal imbalance and the fall in international reserves both in Uruguay and Argentina during the last period of the *tablita* were inconsistent with the *ex ante tablita* rate but were able to explain the *ex post* rate which incorporated the large November 1982 devaluation of the peso.

The cost of imposing monetary discipline through a managed exchange rate system was high due to the fact that people did not expect the monetary authorities to stick to the new policy. As a result, expectations of devaluation (over and above the *tablita* rate) increased domestic interest rates to extraordinarily high levels. The benefits in terms of increased stability, investment, employment, capital inflows and reduced inflation were lost with the peso float of November 1982. The large fiscal imbalance starting in mid-1981 was incompatible with the foreign exchange policy. If the government had to monetize a large fiscal deficit the *tablita* would become unsustainable. This indeed happened in 1982. The policy of managing the exchange rate for stabilization purposes was implemented at a very high initial cost and many of the benefits were lost when the policy was discontinued.

Notes

1. The exchange rate regime was called the '*tablita*' since the government published a 'table' with the daily exchange parities. This regime was adopted in Chile (February 1978), in

Uruguay (October 1978), and in Argentina (December 1978). Mexico has had a similar exchange rate regime for the last few years. In Mexico, however, the government does not publish a table for future exchange parities but at the time the Economic Pact for Solidarity was signed on December 16, 1987, the monetary authorities established a preannounced declining rate of devaluation over time, starting with a devaluation of one peso a day. In December 1988 the new Pact for Economic Growth and Stability was signed and the rate of devaluation was decreased to \$0.80 a day. Since 1989 the devaluation rate has been \$0.40 a day and the government intends to continue this mechanism as an antiinflationary policy up to the time when they are able to fix the exchange rate (up to the time when the rate of devaluation is zero). The idea of fighting inflation with a declining rate of devaluation that would eventually fall to zero was also behind the *tablita* in the Southern Cone economies in the 1970s, except that the only country that moved to a fixed exchange rate system was Chile.

2. While Uruguay completely liberalized its financial sector first and was never able to make great progress in the area of trade liberalization, Chile followed the opposite path by thoroughly liberalizing trade flows and maintaining controls on short-term capita flows. Argentina followed a path somewhere in between by removing most (but not all) restrictions on short-term capital flows before implementing a tariff reform (for details see Corbo *et al.*, 1986).
3. For a detailed analysis of the 1974–82 data see del Castillo (1986).
4. To allow for the existence of frictions arising from transaction costs, information lags, *etc.*, Edwards and Khan (1985) model interest rates in a partial adjustment framework as:

$$i_t = \psi\theta(i_t^* + \delta_t^e) + \psi(1-\theta)i_{t-1} + (1-\psi)(r_t + \pi_t^e),$$

with $0 < \theta < 1$. The faster the adjustments in the financial markets the closer θ will be to 1. Also, the more integrated the domestic economy is (the larger ψ) the more likely the interest rate would adjust more rapidly (the larger θ is).

5. An excess demand (supply) for real money balances will result in a temporarily higher (lower) real interest rate (Mundell's liquidity effect). As Edwards and Khan (1985) point out, *EMS* could also affect inflationary expectations. They also assume that changes in inflationary expectations have no direct effect on real rates of interest (see Mundell, 1963).
6. As Edwards and Khan (1985) point out, an interesting property of their model is that the approximate degree of openness of the financial sector in a particular country can be estimated from the data. They estimated their model for Singapore and Colombia and their results were as expected: in Singapore interest rates were determined by open economy factors only but in Colombia domestic monetary disequilibria and open economy conditions influenced nominal interest rates. As they also point out, one of the limitations of their model is that it assumes a constant degree of openness of the financial sector in the country under study.
7. The problems involved in estimating money demand functions—particularly in developing countries—are well documented in the literature.
8. We could assume either adaptive or rational expectations and the problems involved in doing so are related to what happens when we need to make assumptions about expectations of exchange rate changes, discussed below.
9. In estimations made by Edwards and Khan, the expected rate of devaluation or the forward premium were assumed to be exogenous. As they point out: 'This is quite a restrictive assumption, and a more realistic analysis would have to recognize that the expected exchange rate change is likely to be affected by movements in domestic interest rates and by domestic monetary conditions in general. But recognizing this issue and doing something about it are quite different matters, since *in practice endogenizing the expected rate of devaluation or the forward premium has in most cases proved to be exceedingly difficult*.'
10. Hanson and de Melo indicate that 'The regression equation was not applied to 1982 because of the changes in the economic environment and the apparent change in the constant expected rate of devaluation ...' (sic, p. 925).
11. Hanson and de Melo included a dummy variable to account for the period in which the Central Bank provided low-cost forward cover (second and third quarters of 1981) and

a rough proxy for the real exchange rate. The real exchange rate was found significant in the regression for the period November 1978 to December 1981 but for some unexplained reason was found insignificant if it was extended to the period from January 1978 to December 1981.

12. Blejer and Gil Diaz (1986) developed a model to explain the *ex post* real rate of interest in Uruguay during the period August 1977 to August 1981. Thus they also excluded the period up to November 1982 when the foreign exchange policy was discontinued. They found that the main variables affecting real interest rates in Uruguay were foreign interest rates and changes in the price of traded goods (external factors) and that the magnitude of monetary disequilibrium and changes in the nominal exchange rate (domestic factors) had a minor effect. Blejer and Gil Diaz finding that the real rate of interest in Uruguay was solely determined by external factors is most probably attributed to the way they model risk (as a function of the real exchange rate) and the fact that they do not include in their estimations the last period of the *tablita* (from August 1981 to November 1982) when both policy risk and the real rate were highest.
13. Del Castillo (1988) shows that in the absence of a *tablita*, the actual rate of devaluation is a good descriptor of expectations of devaluation. She shows that for the period 1977:03 to 1978:09 and 1982:12 to 1984:12 a perfect foresight model in which expectations of devaluation are proxied by the actual rate is supported by empirical evidence. On the other hand, she provides empirical evidence to show that when the *tablita* was in operation, the *ex ante* rate of devaluation was an inadequate descriptor of expectations.
14. In Uruguay, for example, the Central Bank only did so from February to September 1981 by offering implicit exchange guarantees. During the first half of 1982 it was the Banco de la República (the state commercial bank) that sold future guarantees.
15. For more evidence on the integration hypothesis see del Castillo (1987). Using the Girton–Roper model of exchange market pressure, adjusted to small open economies by Connolly, the study provides some evidence that a small country may be integrated in such a way that its macroeconomic performance is highly affected by what is happening to its main foreign investors and trading partners in the region—even if those economies are themselves 'small' economies in the world economy. In particular, it is shown that the model performs much better when Argentinian prices (rather than US or Brazilian prices) were used as a proxy for the 'rest of the world' in PPP estimations for Uruguay.
16. The inclusion of Argentine interest rates is justified by the 'small country' assumption. Interest rates in Argentina are an exogenous variable to Uruguay since Uruguayan policy has no impact on Argentinian rates. On the other hand, given that the financial markets are highly integrated, and given the large size of Argentinian flows of capital in and out of the Uruguayan market, Argentine rates can influence interest rates in Uruguay.
17. By analyzing the spread in terms of banks' lending rates (rather than their deposit rates as Hanson and de Melo did) we also abstract from the risk faced by economic agents that their dollar deposits be converted into pesos, as it happened in Mexico in August 1982. Since there is no information on the distribution of loans of different maturities we arbitrarily assumed they were one year loans. Thus, since interest rates in pesos and in dollars are the monthly (annualized) lending rates on loans of up to six months, we calculated the actual *ex post* rate of devaluation as the percentage change in the exchange rate in the following 12 months.
18. The dependent variable is not the difference between peso and dollar interest rates (the spread) but the difference after taking into account the devaluation of the peso (the currency adjusted spread). Given the way we define the actual depreciation of the peso, monthly observations of δ_t^e starting in November 1981 are different *ex ante* and *ex post* since by representing the rate of devaluation in the following 12 months they pick up the effect of the November 1982 devaluation which ended the *tablita* system. During the last year of the *tablita*, those that had borrowed in pesos greatly benefited from the large devaluation of November 1982. If the monetary and fiscal variables that we use as explanatory variables are able to explain the *ex post* but not *ex ante* spread it means that those fiscal and monetary variables are consistent with the devaluation which ended the *tablita* system but were inconsistent—as we claim they would—with the preannounced rate of devaluation as specified in the *tablita*. In the regressions we estimated $D(t)$ is the *ex post* currency adjusted spread, which means that, for the last year, it picks up the effect of the November 1982 devaluation. We also estimated equation 15 with D_t as the

ex ante currency adjusted spread and found that the explanatory variables could not explain the dependent variable. This means that the fiscal disequilibrium and the low level of international reserves were inconsistent with the devaluation rate as specified in the *tablita*.

19. G_t is the ratio of government deficit (IMF, *International Financial* Statistics line 80) as a ratio of GDP. Since there is no available monthly or quarterly data for GDP it had to be interpolated from yearly figures (IMF, *IFS*, line 99b).
20. TRU_t is the percentage changes (d log) in total reserves of Uruguay (total reserves minus gold plus gold at national valuation expressed in millions of SDRs) taken from the IMF, *Supplement on International Reserves*, No. 6, 1983. TRA_t is defined as in TRU_t, for Argentina.
21. $IA(t)$ are annualized active rates of interest (29 days loans) of the Central Bank of Argentina taken from Vatnick (1983).
22. We also estimated the same equation using the preannounced *tablita* rate of devaluation as an additional independent variable (indicated as TAB). As we specified in equation ⟨8⟩, the preannounced rate is one of the components of expectations of devaluation. The following results were obtained

$$\underset{}{\Delta D_t} = \underset{(-0.85)}{-1.69} + \underset{(2.35)}{0.35\Delta G_t} - \underset{(-3.28)}{0.28\Delta TRU_t} - \underset{(-1.83)}{0.12\Delta TRA_t} - \underset{(-2.14)}{0.09\Delta IA_t} + \underset{(2.34)}{0.72 TAB_t},$$

$$R^2 = 0.77 \quad \bar{R}^2 = 0.73 \quad \mathrm{DW} = 2.03 \quad F = 19.86 \quad AR = 2$$

All variables remain highly significant and the t-statistics of the TAB coefficient are significant at the 2 per cent level. We thus confirm that economic agents take into account the preannounced rate of devaluation when they make the decision on whether to borrow in pesos or dollars. When used alone, however, this variable is much less successful in explaining the *ex post* currently adjusted spread than when used with the fiscal and monetary variables specified in the model.
23. The need to correct for second-order autocorrelation is warranted by the moving average representation of the variables. Since the errors were serially correlated and not white noise, GLS was used to correct for serial correlation.
24. See note 18.
25. While the first three coefficients are significant at the 1 per cent level, the fourth coefficient is significant at the 2 per cent level.

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